BCA GENERAL — COURSE OUTLINE

Third Year • Semesters V & VI • 2024 admission batch

Important: This pack is prepared on the assumption that your college is affiliated to **Bangalore University (BU)**, not Bengaluru City University (BCU). The BU BCA NEP 2021 scheme document states it applies from academic year 2021–22 onwards. Confirm your scheme and college notices before relying on dates or elective choices.

Basis and scope

The subject list below is transcribed from the Bangalore University BCA NEP 2021 scheme (academic year 2021–22 onwards), which is the identified BU scheme document relevant to a 2024 entrant. This is a concise course outline, not a replacement for the complete university syllabus or your college's approved elective selection.

Semester V

Course	Credits	Main coverage / learning focus
Artificial Intelligence (CA-C21T)	4	Intelligent agents; problem-solving search (DFS, BFS, A*, AO*); game search; logic and knowledge representation; planning; uncertainty and fuzzy logic; robotics; computer vision; NLP; expert systems; machine learning and neural networks.
Data Analytics (CA-C22T)	4	Analytics types and applications; probability and statistical methods; hypothesis testing, correlation and regression; data visualization; Power BI; business case studies.
Web Programming (CA-C23T)	4	Internet and web fundamentals; JavaScript and DOM; events; XML and web services; PHP syntax, functions, arrays, file handling, exceptions, cookies and database connectivity.
Data Analytics Lab (CA-C24L)	2	Spreadsheet analysis and dashboards; Python probability/statistics, tests, correlation and regression; Power BI reports and dashboards.
Web Programming Lab (CA-C25L)	2	HTML forms and validation; JavaScript calculations and dynamic effects; PHP programs, database operations, cookies, file handling and input validation.
Vocation Course I: Quantitative Techniques (CA-V1)	3	Numerical ability, probability, reasoning, time/work/distance, data interpretation, arithmetic, communication and research/teaching aptitude topics as set out in the scheme.
Elective I (CA-E1)	3	Choose the college/university-approved option: Data Mining OR Computer Graphics. Confirm the option actually offered to your batch.
SEC III	—	Cyber Crime, Cyber Law and Intellectual Property Right.

Semester VI

Course	Credits	Main coverage / learning focus
Project Work (CA-C26P)	4	Applied project: problem definition, literature/background study, requirements, design, implementation, testing, documentation and presentation/viva as directed by the department.
Machine Learning (CA-C27T)	4	Machine-learning foundations and methods. Refer to the full BU syllabus for unit-wise topics, prescribed texts and assessment details.
Mobile Application Development (CA-C28T)	4	Mobile application concepts, development environment, interface design, application components, data handling and deployment; verify exact unit breakdown in the full syllabus.
Machine Learning Lab (CA-C29L)	2	Practical exercises and implementation work aligned to the Machine Learning paper.

Course	Credits	Main coverage / learning focus
Mobile Application Development Lab (CA-C30L)	2	Practical mobile application development exercises aligned to the theory paper.
Vocation Course II: Electronic Content Design (CA-V2)	3	Electronic/digital content design course as listed in the BU scheme.
Elective II (CA-E2)	3	Choose the college/university-approved option: Operations Research OR Software Testing. Confirm the option offered to your batch.
Internship	2	Internship component; follow department instructions for duration, logbook, certificate and evaluation.

Scheme notes

The BU scheme table lists 25 total credits for Semester V and 24 total credits for Semester VI. The complete official syllabus contains unit-wise descriptions, practical lists, textbooks and assessment scheme; use it for detailed preparation. Elective availability and exact course implementation can be college-specific.

Official/reference sources

• BU BCA NEP 2021 scheme, V & VI semesters (PDF): https://www.suranacollegepeenya.edu.in/wp-content/uploads/2023/11/BCA-NEP-5TH-AND-6TH.pdf
• Bangalore University official website: https://b-u.ac.in/